Radxa CM4 on DF40C sockets

Pin allocation: docs/head-pcb.md section 4. Unused pins not connected, as on the NANO-A

J9

GND	2	1	GND
	Ethernet_Pair1_P	Ethernet_Pair3_P	
	Ethernet_Pair1_N	Ethernet_Pair3_N	
GND	6	5	GND
	Ethernet_Pair0_N	Ethernet_Pair2_N	
	Ethernet_Pair0_P	Ethernet_Pair2_P	
GND	10	9	GND
	Ethernet_SYNC_IN(1.8v)	Ethernet_nLED3(3.3v)	
	Ethernet_SYNC_OUT(1.8v)	Ethernet_nLED2(3.3v)	
	EEPROM_nWP	Ethernet_nLED1(3.3v)	
GND	14	13	GND
SAI2_LRCK	GPIO26	GPIO21	SAI2_SDO
SPI2_MOSI	GPIO19	GPIO20	SAI2_SDI
SPI_INT	GPIO13	GPIO16	
GND	GPIO6	GPIO12	SPI2_CLK
SPI_IO	GPIO5	GND	GND
CAM_SDA	ID_SD	ID_SC	CAM_SCL
	GPIO11	GPIO7	
	GPIO9	GPIO8	
GND	GND	GPIO25	
EYE_DATA	GPIO10	GND	GND
SPI2_MISO	GPIO22	GPIO24	UART7_RX
	GPIO27	GPIO23	UART7_TX
FAN_PWM	GPIO17	GPIO18	SAI2_SCLK
GND	GND	GPIO15	
SPI2_CS0	GPIO4	GND	GND
I2C8_SCL	GPIO3	GPIO14	
I2C8_SDA	GPIO2	SD_CLK	
GND	GND	GND	GND
	SD_CMD	SD_DAT3	
	SD_DAT5	SD_DAT0	
GND	GND	GND	GND
	SD_DAT4	SD_DAT1	
	SD_DAT7	SD_DAT2	
	SD_DAT6	GND	GND
GND	GND	SD_VDD_Override	
	Reserved	SD_PWR_ON	
CM4_3V3	GPIO_VREF(1.8v/3.3v_Input)	+5v_(Input)	+5V
	SCL0	+5v_(Input)	+5V
	SDA0	+5v_(Input)	+5V
CM4_3V3	+3.3v_(Output) 600mA Max	+5v_(Input)	+5V
	+3.3v_(Output)	+5v_(Input)	+5V
	+1.8v_(Output) 600mA Max	+5v_(Input)	+5V
	+1.8v_(Output)	WiFi_nDisable	
CM4_RESET	RUN_PG	BT_nDisable	
	AnalogIP1	nRPIBOOT	CM4_MASKROM
	AnalogIP0	nPI_LED_PWR	
GND	GND	Camera_GPIO	CAM_GPIO
	nEXTRST	Global_EN	

NB SD signals are only available on modules without eMMC

CM4 socket 1–100

J10

	102	101	
	PCIe_CLK_nREQ	USB_OTG_ID	
	104	103	USB_DN
	Reserved	USB2_N	
	106	105	USB_DP
GND	Reserved	USB2_P	
	108	107	GND
	GND	PCIe_nRST	
	110	109	
	PCIe_CLK_P	VDAC_COMP	
	112	111	
GND	PCIe_CLK_N	GND	
	114	113	GND
	GND	CAM1_D0_N	
	116	115	
	PCIe_RX_P	CAM1_D0_P	
	118	117	
GND	PCIe_RX_N	GND	GND
	120	119	
	GND	CAM1_D1_N	
	122	121	
	PCIe_TX_P	CAM1_D1_P	
	124	123	
	PCIe_TX_N	GND	GND
GND	126	125	
CSI_D0_N	GND	CAM1_C_N	
CSI_D0_P	128	127	
	CAM0_D0_N	CAM1_C_P	
GND	130	129	
CSI_D1_N	CAM0_D0_P	GND	GND
CSI_D1_P	132	131	
	GND	CAM1_D2_N	
	134	133	
GND	CAM0_D1_N	CAM1_D2_P	
CSI_C_N	136	135	
CSI_C_P	CAM0_D1_P	GND	GND
	138	137	
	GND	CAM1_D3_N	
	140	139	
	CAM0_C_N	CAM1_D3_P	
GND	142	141	
	CAM0_C_P	HDMI1_HOTPLUG	
	144	143	
	GND	HDMI1_SDA	
	146	145	
	HDMI1_TX2_P	HDMI1_SCL	
	148	147	
GND	HDMI1_TX2_N	HDMI1_CEC	
	150	149	
	GND	HDMI0_CEC	
	152	151	
	HDMI1_TX1_P	HDMI0_HOTPLUG	
	154	153	
GND	HDMI1_TX1_N	GND	GND
	156	155	
	GND	DSI0_D0_N	
	158	157	
	HDMI1_TX0_P	DSI0_D0_P	
	160	159	
GND	HDMI1_TX0_N	GND	GND
	162	161	
	GND	DSI0_D1_N	
	164	163	
	HDMI1_CLK_P	DSI0_D1_P	
	166	165	
	HDMI1_CLK_N	GND	
GND	168	167	GND
	GND	DSI0_C_N	
	170	169	
	HDMI0_TX2_P	DSI0_C_P	
	172	171	
GND	HDMI0_TX2_N	GND	GND
	174	173	
	GND	DSI1_D0_N	
	176	175	
	HDMI0_TX1_P	DSI1_D0_P	
	178	177	
	HDMI0_TX1_N	GND	
GND	180	179	GND
	GND	DSI1_D1_N	
	182	181	
	HDMI0_TX0_P	DSI1_D1_P	
	184	183	
	HDMI0_TX0_N	GND	
GND	186	185	GND
	GND	DSI1_C_N	
	188	187	
	HDMI0_CLK_P	DSI1_C_P	
	190	189	
	HDMI0_CLK_N	GND	
GND	192	191	GND
	DSI1_D3_N	DSI1_D2_N	
	194	193	
	DSI1_D3_P	DSI1_D2_P	
	196	195	
GND	GND	GND	GND
	198	197	
	HDMI0_SCL	HDMI0_SDA	
	200	199	

High Speed Serial

CM4 socket 101–200

GPIO_VREF (78) on the module's own 3.3 V output (84; 86 is the same rail, left open): 3.3 V GPIO, as the NANO-A's H2 jumper

H5 — Radxa CM4

Head bundle in, 5 V buck

LMR51635Y, 1.1 MHz, 4.7 uH (Ti table 8-2: 3.3 uH), 82k/15k sets 5.17 V, EN tied to VIN. F6 (2 A) on the bridge fuses this feed.
JP1 joins the buck to the 5 V rail: cut it to feed 5 V from outside (J12). The feedback divider stays on the buck side

J1 — VBAT_IN 1, GND 2, LINK_RX 3, LINK_TX 4 — Head bundle 1 +BATT, 2 GND, 3 bridge TX, 4 bridge RX (bridge J6 order)

U2 LMR51635YDDCR: VIN 3, EN 4, CB 6 BUCK_CB, SW 5 BUCK_SW, FB 1 BUCK_FB, GND

C5 BUCK_CB–BUCK_SW 100n 50V

L1 BUCK_SW–5V_BUCK 4.7u

R1 5V_BUCK–BUCK_FB 82k

R2 BUCK_FB–GND 15k

C6 5V_BUCK–GND 47u 10V

C7 5V_BUCK–GND 47u 10V

C8 5V_BUCK–GND 47u 10V

JP1 5V_BUCK–+5V — cut for ext 5V

PWR_FLAG +5V

D1 SMBJ30A VBAT_IN–GND

C1 10u 50V X7R VBAT_IN–GND

C2 10u 50V X7R VBAT_IN–GND

C17 47u 50V VBAT_IN–GND

C4 100n 50V VBAT_IN–GND

PWR_FLAG VBAT_IN

PWR_FLAG GND

Fan and eye LED

Fan: low side, PWM1_CH2 (pin 50). Eye: WS2812-type, SPI1 MOSI (pin 44), 3.3 V data

D2 B5819W +5V–FAN_N

J6 +5V, FAN_N — Fan 1 +5V, 2 FAN−

R5 FAN_PWM–FAN_G 100R

Q1 2N7002 FAN_G, FAN_N, GND

R6 FAN_G–GND 100k

J4 +5V, EYE_DATA, EYE_DIN, GND — Eye LED 1 +5V, 2 DATA, 3 GND

R7 EYE_DATA–EYE_DIN 330R

I2C8: head IMU, ToF, breakout

BMI088 on the board. I2C: PS and CSB1 high, CSB2 open (allowed in I2C mode): SDO1 and SDO2 high: accelerometer 0x19, gyroscope 0x69 (strapped so the pin rows need no vias between alternate pins).
R8, R9: the bus pull-ups (the ToF breakout adds 10k). Cut JP4 and JP5 to use an external IMU module on J3

U5 BMI088: +3V3, VDD, VDDIO, CSB1, CSB2, SCK/SCL, SDI/SDA, SDO1, SDO2, PS, GNDA, GNDIO, INT1–INT4; GND GND

R8 +3V3–I2C8_SDA 4,7k

R9 +3V3–I2C8_SCL 4,7k

J2 — GND, +3V3, I2C8_SDA, I2C8_SCL — ToF VL53L8CX breakout, Qwiic order

C15 +3V3–GND 100n 50V

C16 +3V3–GND 100n 50V

C15 at VDD, C16 at VDDIO

J3 — GND, +3V3, I2C8_SDA, I2C8_SCL — I2C breakout. Spare I2C8, same order: external IMU module here

JP4 I2C8_SDA–IMU_SDA — cut for ext IMU

JP5 I2C8_SCL–IMU_SCL — cut for ext IMU

Power LED, mounting holes

M2 holes, one per corner, not connected

R12 +5V–LED_A 1k

D3 LED_A–GND green

H1 M2

H2 M2

H3 M2

H4 M2

SPI breakout

SPI2 M1 for a future SPI device, untested. Wire pads on the bottom, under the module, along the top edge.
CLK GPIO12 (CM4 31), MOSI GPIO13 (28), MISO GPIO22 (46), CS0 GPIO4 (54), INT GPIO6 (30), spare GPIO5 (34)

J13 — GND, +5V, +3V3, SPI2_MOSI, SPI2_MISO, SPI2_CLK, SPI2_CS0, SPI_INT, SPI_IO — SPI pads — J13: 1 GND, 2 5V, 3 3V3, 4 MOSI, 5 MISO, 6 CLK, 7 CS0, 8 INT, 9 spare GPIO

3.3 V

AMS1117-3.3 from 5 V, as on the NANO-A. Camera, ToF, IMU, mic, I2C breakout

U3 AMS1117-3.3: +5V, VI, VO, GND, +3V3

C9 +5V–GND 10u 25V

C10 +3V3–GND 22u 25V

C11 +3V3–GND 100n 50V

Speaker amplifier

MAX98357A on SAI2. SD_MODE 1M to 5 V: (L+R)/2 mono, GAIN open = 9 dB: fit R4 (100k to GND) for 15 dB.
JP3 feeds it: cut JP3 to use an external amplifier module on J11

JP3 +5V–AMP_5V — cut for ext amp

U4 MAX98357AETE+T: VDD, DIN, BCLK, LRCLK, GAIN_SLOT, SD_MODE, OUTP, OUTN, GND, PAD

SAI2_SDO 1, SAI2_SCLK 16, SAI2_LRCK 14

AMP_SD 4, AMP_GAIN 2

C12 AMP_5V–GND 10u 25V

C13 AMP_5V–GND 100n 50V

PWR_FLAG AMP_5V

J5 SPK_P 1, SPK_N 2 — Speaker

Bridge-tied output: neither lead to GND

R3 AMP_5V–AMP_SD 1M

R4 AMP_GAIN–GND 100k

Microphone

INMP441 on SAI2. L/R tied to GND on the mic side (left slot)

J7 — +3V3, GND, SAI2_SCLK, SAI2_LRCK, SAI2_SDI — Microphone 1 3V3, 2 GND, 3 SCK, 4 WS, 5 SD

Camera FPC

Pi 15-pin camera order. CAM0 lanes (Radxa CSI1), I2C6 with 4.7k pull-ups, CAM_GPIO enable. 100 ohm pairs

J8 — GND, CSI_D0_N, CSI_D0_P, GND, CSI_D1_N, CSI_D1_P, GND, CSI_C_N, CSI_C_P, GND, CAM_GPIO, CAM_SCL, CAM_SDA, +3V3 — Camera

R10 +3V3–CAM_SCL 4,7k

R11 +3V3–CAM_SDA 4,7k

C14 +3V3–GND 100n 50V

Pin 12 (camera clock) not connected: the module has its own oscillator

External fallbacks

Wire pads, 2.54 mm. J12: cut JP1 and feed 5 V from an external step-down. J11: cut JP3 and wire an amplifier module, pins in the generic module's order. J3 (section 6) takes an external IMU module

J12 — VBAT_IN, GND, +5V, GND — Ext 5V — J12: 1 B+ out, 2 GND, 3 5V in, 4 GND

J11 — SAI2_LRCK, SAI2_SCLK, SAI2_SDO, GND, +5V — Ext amp — J11: LRC, BCLK, DIN, GAIN (open), SD (open: module's own pull-up), GND, Vin

Link series resistors, test pads

330R in each UART7 line limits current into an unpowered CM4 from the bridge. TP1/TP2 (bottom): short to TP3 to reset: hold TP2 low through a reset for maskrom. TP4–TP6 (bottom): USB D−, D+, GND of the download port, for rkdeveloptool in maskrom

R13 UART7_RX–LINK_RX 330R

R14 UART7_TX–LINK_TX 330R

TP1 CM4_RESET

TP2 CM4_MASKROM

TP3 GND

TP4 USB_DN

TP5 USB_DP

TP6 GND

Bootstrapped by scripts/kicad/gen_head_sch.py; edit this file from now on
Design note: docs/head-pcb.md
File: head.kicad_sch

Title: Dukki head carrier

Size: A2 | Date: 16/10/2026 | KiCad E.D.A. 10.0.0 | **Rev: B** | Id: 1/1